Botao He

botao@umd.edu · botao.me · Google Scholar

Research Profile

I build robots that learn through physical interaction, and real-world systems that make that learning work.

Education

University of Maryland, College Park — Ph.D. in Computer Science Aug. 2022–present
Perception and Robotics Group; advised by Yiannis Aloimonos and Cornelia Fermüller.

Nanjing Institute of Technology — B.Eng. in Robotics Sep. 2018–Jul. 2022

Research & Industry Experience

Amazon Frontier AI & Robotics Team — Research Intern Jan. 2026–present

- Large-scale contact-aware data processing for robot video world model pretraining.
- Developing force-aware dexterous manipulation methods from multimodal human interaction data.
- Exploring physics-grounded, interaction-aware video and world models for robot learning.

Honda Research Institute USA — Research Intern May–Aug. 2025
Developed robust motion prediction using a trainable mixture of model-based and learned experts.

Carnegie Mellon University — Research collaboration May 2023–2025
Interactive navigation and field autonomy; collaborated with Ji Zhang.

Selected Research & Systems

PlayPolicy — dexterity from supervised play
Bootstraps a goal-agnostic play policy from minutes of human play to collect hours of contact-rich data. Play-policy-guided MPC achieves new goals without task-specific demonstrations.

DreamFAR — real-time, long-horizon world simulator **Project Page**
Pre-trained from scratch on robot teleoperation data, with contact-anchored memory that keeps minutes-long autoregressive rollouts consistent.

CARD — contact-anchored dexterous retargeting
Real-time human-to-robot retargeting without object models or simulation, preserving contact and force alongside kinematics.

ForceBand — bring force into human data with EMG, forceful manipulation directly from human data
ForceBand is a low-cost, wrist-worn sEMG band that predicts fingertip forces from forearm muscle activity, enabling force-enriched human demonstrations for learning forceful robot manipulation—without any force sensors on the hand.

FEEL — force-enhanced egocentric dataset ECCV 2026 Long Oral (25/10473)
A large-scale dataset pairing egocentric video with calibrated glove-based force measurements, comprising approximately three million synchronized frames of natural manipulation.

Interactive-FAR & Air-FAR — deployable field autonomy
Built navigation systems for ground and aerial robots operating in complex unknown environments. Interactive-FAR reasons about movable obstacles online; Air-FAR extends fast, adaptable routing to large-scale aerial navigation.

Microsaccade-Inspired Event Camera — sensing hardware and algorithms
Co-designed an actuated event-camera mechanism and computational pipeline that reveals static scene structure while preserving compatibility with established event-based perception methods.

Selected Publications

- **Botao He**, Yiming Dou, Yiannis Aloimonos, Jitendra Malik, Akash Sharma, Ruoshi Liu. *PlayPolicy: Dexterity from Supervised Play.* arXiv:2610.22340, 2026.
- **Botao He**, Zhi Wang, Linna Kuang, Ishaan Ghosh, Jitendra Malik, Cornelia Fermüller, Tingfan Wu, Jiayuan Mao, Ruoshi Liu, Haozhi Qi, Yiannis Aloimonos. *ForceBand: Learning Forceful Manipulation with sEMG.* CoRL 2026.
- Yiming Dou, **Botao He**, Antonio Loquercio, Andrew Owens, Jitendra Malik, Sumedh Anand Sontakke, Ruoshi Liu. *DreamFAR: Real-Time & Long-Horizon Interaction with Diverse Objects.* Preprint, 2026.
- Cunxi Dai, **Botao He**, Francois R. Hogan, Wei Pan, et al. *CARD: Contact-Anchored Retargeting for Dexterous Manipulation.* arXiv:2610.36539, 2026.
- Eadom Dessalene, **Botao He**, Michael Maynord, et al. *FEEL (Force-Enhanced Egocentric Learning): A Dataset for Physical Action Understanding.* ECCV 2026, **Long Oral (25/10,473 submissions)**.
- Zhi Wang, **Botao He**, Kelin Yu, et al. *HumanEgo: Zero-Shot Robot Learning from Minutes of Human Egocentric Videos.* IROS 2026 WORLDS Workshop **Best Paper**.
- Yu Chen, **Botao He**, Yuemin Mao, et al. *Adversarial Game-Theoretic Algorithm for Dexterous Grasp Synthesis.* Preprint, 2025.
- **Botao He***, Amir Hossein Shahidzadeh*, Yu Chen*, et al. *NavMoE: Hybrid Model- and Learning-based Traversability Estimation for Local Navigation via Mixture of Experts.* ICRA, 2025.
- **Botao He**, Guofei Chen, Cornelia Fermüller, Yiannis Aloimonos, Ji Zhang. *Air-FAR: Fast and Adaptable Routing for Aerial Navigation in Large-scale Complex Unknown Environments.* ICRA, 2025.
- **Botao He***, Guofei Chen*, Wenshan Wang, et al. *Interactive-FAR: Interactive, Fast and Adaptable Routing for Navigation among Movable Obstacles in Complex Unknown Environments.* IROS, 2024.
- **Botao He**, Ze Wang, Yuan Zhou, et al. *Microsaccade-Inspired Event Camera for Robotics. Science Robotics*, 2024.
- Qianhao Wang*, **Botao He***, Zhiren Xun, Chao Xu, Fei Gao. *GPA-Teleoperation: Gaze Enhanced Perception-aware Safe Assistive Aerial Teleoperation.* IEEE Robotics and Automation Letters 7(2), 2022; presented at ICRA 2022.
- **Botao He***, Haojia Li*, Siyuan Wu, et al. *FAST-Dynamic-Vision: Detection and Tracking Dynamic Objects with Event and Depth Sensing.* IROS, 2021.

** Equal contribution. Full publication record:* Google Scholar.

Technical Skills

Learning	Imitation learning, world models, diffusion policies, multimodal learning, online adaptation
Robotics	Dexterous manipulation, motion planning, navigation, active perception, ROS, Gazebo, AirSim, Unity
Programming	Python, C/C++, MATLAB, Git, Linux
Hardware	Custom sensing, embedded systems, STM32/Arduino, SolidWorks, machining, circuit design

Service & Recognition

Reviewing	Science Robotics; IEEE TPAMI; IEEE RA-L; IEEE T-ITS; ICRA; IROS; Frontiers in Robotics and AI
Recognition	NeuroPAC Fellowship (2023, 2024); University of Maryland Dean's Fellowship (2022–2023)